

Le Zheng

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RESEARCH INTERESTS

Dexterous Manipulation | Mobile Manipulation | End-to-End Navigation

EDUCATION

Sun Yat-sen University

Master of Computer Technology

- Advisor: Prof. Hui Cheng, Rapid Lab
- Admission: Recommended for admission without examination
- Average Score: 89/100

Guangzhou, China

Sep 2024 — Jul 2027

Northeastern University

Bachelor of Measurement and Control Technology and Instrument

- Average Score: 87/100

Shenyang, China

Sep 2020 — Jul 2024

WORK EXPERIENCE

Embodied AI Algorithm Intern

Agibot

- Constructed and integrated the multimodal **Vision-Tactile-Language-Action (VTLA)** algorithm architecture for bimanual dexterous manipulation tasks (earphone flipping / multi-object grasping).
- Trained $\pi_{0.5}$ models and conducted **Visual-Tactile Joint Pre-training**. Streamlined the full pipeline from high-quality multimodal data collection and cleaning to large model pre-training and post-training, successfully deploying it on real dexterous hands.
- Co-authored **STAR**, currently under review at **ICRA 2027**.

Mar 2026 — Jul 2026

Shanghai, China

Research Intern

Rapid Lab, Sun Yat-sen University

- Designed and implemented visual servo reinforcement learning algorithms targeting **mobile manipulation** tasks (door opening / grasping).
- Developed PPO reinforcement learning algorithms based on the **iGibson** simulator and designed task-oriented reward functions for end-effector decision-making.
- Co-authored related research accepted by **ICRA 2025**.

Apr 2024 — Aug 2024

Guangzhou, China

Algorithm Group Leader

T-DT Lab, Northeastern University

- Led the algorithm team in driving technical breakthroughs and tactical coordination.
- Architected and deployed visual recognition algorithms and moving target tracking strategies, significantly enhancing the robot's auto-aiming accuracy in real combat scenarios. Designed automatic infantry decision algorithms for the 2023 season.
- Achieved **Regional Champion** in the 2022 season and **National Runner-up** in the 2023 season of the RoboMaster University Championship (RMUC).

Sep 2021 — Aug 2023

Shenyang, China

PUBLICATIONS

Under Review

- **GeoPT: Privileged Geometric Post-Training for One-Step Whole-Body Motion Planning in Mobile Manipulation**

Le Zheng, Shuhang Wang, Zixuan Zhuang, Hao Ren, Yiming Zeng, Hui Cheng

IEEE International Conference on Robotics & Automation (ICRA), 2027 | [Website](#) | [Paper](#)

- **STAR: Sparse Tactile Representation Learning in Vision-Tactile-Language-Action Models for Dexterous Manipulation**

Xiangcheng Liu*, Tianhao Wu*, **Le Zheng***, Yidong Wang, Bowen Jiang, Mingjie Pan, Xinlin Ren, Yi Liu, Jianlan Luo
IEEE International Conference on Robotics & Automation (ICRA), 2027 | [Website](#) | [arXiv](#)

Published

- **Unordered Landmark Visual Navigation**

Hao Ren, Junzhe Zhu, Yihan Li, Zetong Bi, **Le Zheng**, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Dizhe Zhang, Lu Qi, Hui Cheng

The 19th European Conference on Computer Vision (ECCV), 2026 | **Oral** | [Website](#) | [arXiv](#)

- **Fisher-Preserving Guidance: Training-Free Manifold Constraints for Safe Diffusion Control**

Hao Ren, Zetong Bi, Yiming Zeng, **Le Zheng**, Zhi Li, Zhaoliang Wan, Lu Qi, Hui Cheng

Proceedings of the 43rd International Conference on Machine Learning (ICML), 2026 | **Poster** | [arXiv](#)

- **SR-Planner: Sampling-Based Path Planning with Feasibility-Aware Focus Regions and Robust Trajectory Optimization for Mobile Manipulator**

Shuhang Wang, **Le Zheng**, Yiming Zeng, Hui Cheng

IEEE Robotics and Automation Letters (R-AL), 2025 | [Paper](#) | [Video](#)

- **RM-Planner: Integrating Reinforcement Learning with Whole-Body Model Predictive Control for Mobile Manipulation**

Zixuan Zhuang*, **Le Zheng***, Wanyue Li, Renming Liu, Peng Lu, Hui Cheng

IEEE International Conference on Robotics & Automation (ICRA), 2025 | [Paper](#) | [Video](#)

* Equal contribution.

SELECTED ACADEMIC PROJECT

Diff-Planner: Cascaded Diffusion-Based Quadrotor Navigation, (2025)

- Proposed a cascaded diffusion network integrating expert trajectory approximation and safety-guided denoising constraints.
- Designed an end-to-end UAV planning framework without explicit mapping, achieving closed-loop navigation based solely on 96x96 depth images.
- Validated the system through real-world integration and deployment on a 3.5-inch quadrotor equipped with Jetson Orin NX, RealSense D435i, and PX4 flight controller.

SKILLS

- **Robotics & Real-world Deployment:** Experienced in hardware debugging and physical deployment of multimodal dexterous hands, bimanual manipulation systems, mobile manipulators, and UAVs. Proficient in ROS/ROS 2; mastered NVIDIA Isaac ecosystem (Isaac Sim, Isaac Lab) and simulation/planning tools like CuRobo, Flightmare, and MarSim; familiar with NVIDIA Jetson (Orin) hardware tuning.
- **Algorithms:** Proficient in PyTorch/JAX and OpenCV. Strong capability in Generative Models (DDPM, EDM, Flow Matching) and Reinforcement Learning (Standard RL: PPO, Q-Learning; Post-Training RL: DPPO, DTQL). Deep understanding of $\pi_{0.5}$ VTLA multimodal alignment, pre-training, and post-training pipelines.

AWARDS

- **ICM 2023:** Finalist (Top 3%) in the Interdisciplinary Contest in Modeling (ICM).
- **RMUC 2023:** National Runner-up (2nd Place) in DJI RoboMaster University Championship (RMUC).